Radiation reaction

$$\frac{dp^i}{ds} = -\frac{\mu_0 q^2 c}{6\pi} \frac{du^k}{ds} \frac{du_k}{ds} u^i$$

rate of change of 4-momentum of radiation

Manipulate it: $u_k u^k = 1 \Rightarrow \frac{du_k}{ds} u^k + u_k \frac{du^k}{ds} = 0$

$$\Rightarrow u_k \frac{du^k}{ds} = 0$$

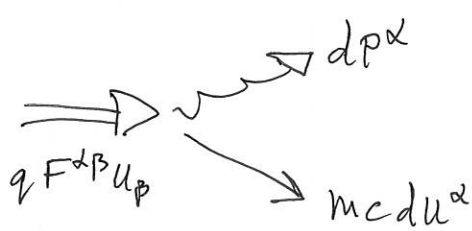
$$\Rightarrow \frac{du_k}{ds} \frac{du^k}{ds} + u_k \frac{d^2 u^k}{ds^2} = 0$$

$$\Rightarrow \frac{du^k}{ds} \frac{du_k}{ds} = -u^k \frac{d^2 u_k}{ds^2}$$

$$\Rightarrow \frac{dp^i}{ds} = \frac{\mu_0 q^2 c}{6\pi} \frac{d^2 u_k}{ds^2} u^k u^i$$

We would like to argue now that the <u>total</u> 4-momentum change of particle + radiation system is

$$\underbrace{mc \frac{du^\alpha}{ds} + \frac{dp^\alpha}{ds}}_{\frac{d}{ds}\text{ total momentum}} = q F^{\alpha\beta} u_\beta \quad \text{(Lorentz force)}$$



$$mc \frac{du^\alpha}{ds} + \frac{\mu_0 q^2 c}{6\pi} \frac{d^2 u_\beta}{ds^2} u^\beta u^\alpha = q F^{\alpha\beta} u_\beta$$

However this can not be right because if we multiply both sides by u_α we get:

$$mc \overbrace{u_\alpha \frac{du^\alpha}{ds}}^{0} + \frac{\mu_0 q^2 c}{6\pi} \frac{d^2 u_\beta}{ds^2} u^\beta \underbrace{u^\alpha u_\alpha}_{1} = q \overset{\text{antisymmetric}}{F^{\alpha\beta}} u_\beta u_\alpha = 0$$

$\Rightarrow$ we are left with $\frac{\mu_0 q^2 c}{6\pi} \frac{d^2 u_\beta}{ds^2} u^\beta$ on the LHS.

"Fix" equation by replacing $\frac{d^2 u_\beta}{ds^2} u^\beta u^\alpha$ with

$$\left(\frac{d^2 u_\beta}{ds^2} u^\beta u^\alpha - \frac{d^2 u^\alpha}{ds^2} \right)$$

The "fixed" equation becomes

$$mc \frac{du^\alpha}{ds} = q F^{\alpha\beta} u_\beta + \frac{\mu_0 q^2 c}{6\pi} \left(\frac{d^2 u^\alpha}{ds^2} - \frac{d^2 u_\beta}{ds^2} u^\beta u^\alpha \right)$$

(Dirac eqn. of motion)

let's look on its non-relativistic limit: $\gamma \to 1$

$u^\alpha \to (1, \vec{v}/c) \qquad ds \to c\,d\tau \to c\,dt$

spatial part

$$mc \frac{d}{c\,dt}\left(\frac{\vec{v}}{c}\right) = \frac{q}{c}\left(\vec{E} + \vec{v} \times \vec{B}\right) + \frac{\mu_0 q^2 c}{6\pi}\left\{ \frac{1}{c^2} \frac{d^2}{dt^2}\left(\frac{\vec{v}}{c}\right) - \frac{1}{c^2} \frac{d^2}{dt^2}\left(\frac{\vec{v}}{c}\right) \cdot \frac{\vec{v}}{c} \left[\frac{\vec{v}}{c}\right] \right\}$$

higher powers of $\frac{\vec{v}}{c}$ $\Rightarrow$ drop it.

$$m \frac{d\vec{v}}{dt} = q\left(\vec{E} + \vec{v} \times \vec{B}\right) + \frac{\mu_0 q^2}{6\pi c} \frac{d^2 \vec{v}}{dt^2}$$ (Abraham-Lorentz eqn)

"Radiation friction" Non-relativistic limit

Power emitted by moving particle

$$\underline{P} = \int I \, d\Omega = \frac{\mu_0 q^2}{6\pi c} \dot{v}^2 = \text{rate of energy loss}$$

$\Rightarrow$ there must be a force $\vec{F}_{rad}$ acting on q

Work by it $\int_0^{t_1} \vec{F}_{rad} \cdot \vec{v} \, dt = -\frac{\mu_0 q^2}{6\pi c} \int_0^{t_1} dt \, \dot{v}^2$

Here we make big assumption: charge is performing stationary motion (aside from P being produced) so that $-\int P \, dt$ represents <u>all</u> energy lost in time t_1

$$\int \dot{\vec{v}} \cdot \dot{\vec{v}} \, dt = \int \frac{d\vec{v}}{dt} \frac{d\vec{v}}{dt} dt = \int_0^{t_1} \frac{d\vec{v}}{dt} d\vec{v} =$$

$$= \underbrace{\left(\vec{v} \cdot \frac{d\vec{v}}{dt} \right) \Big|_0^{t_1}}_{=0} - \int_0^{t_1} \vec{v} \frac{d^2\vec{v}}{dt^2} dt = -\int \vec{v} \cdot \frac{d\vec{a}}{dt} dt$$

$$\Rightarrow \int_0^{t_1} \left(\vec{F}_{rad} - \frac{\mu_0 q^2}{6\pi c} \dot{\vec{a}} \right) \vec{v} \, dt = 0$$

radiation reaction force (radiation friction)

$$\vec{F}_{rad} = \frac{\mu_0 q^2}{6\pi c} \frac{d\vec{a}}{dt} \quad \text{Abraham - Lorentz f-la}$$

$$= \frac{\mu_0 q^2}{6\pi c} \ddot{\vec{v}}$$

<u>Pathology</u>: $m \dot{\vec{v}} = \frac{\mu_0 q^2}{6\pi c} \ddot{\vec{v}} \Rightarrow \vec{v} = \text{const}$

free-particle

OR

$$\vec{v} = \vec{v}_0 e^{\alpha t}$$

$$m \vec{v}_0 \alpha e^{\alpha t} = \frac{\mu_0 q^2}{6\pi c} \vec{v}_0 \alpha^2 e^{\alpha t}$$

Runaway solutions of Abraham-Lorentz equ:

$$m\frac{d\vec{v}}{dt} = \underset{\substack{\uparrow \\ \text{Lorentz force}}}{\vec{F}(t)} + m\tau_e \frac{d^2\vec{v}}{dt^2} \quad , \quad \alpha = m\tau_e = \frac{\mu_0 q^2}{6\pi c}$$

$$\frac{d\vec{v}}{dt} = \frac{\vec{F}(t)}{m} + \tau_e \frac{d^2\vec{v}}{dt^2}$$

$$\frac{d^2\vec{v}}{dt^2} - \frac{1}{\tau_e}\frac{d\vec{v}}{dt} = -\frac{1}{m\tau_e}\vec{F}(t)$$

let $$\frac{d^2\vec{v}}{dt^2}e^{-t/\tau_e} - \frac{1}{\tau_e}\frac{d\vec{v}}{dt}e^{-t/\tau_e} = -\frac{1}{m\tau_e}\vec{F}(t)e^{-t/\tau_e}$$

But $$\frac{d}{dt}\left(\frac{d\vec{v}}{dt}e^{-t/\tau_e}\right) = \frac{d^2\vec{v}}{dt^2}e^{-t/\tau_e} - \frac{1}{\tau_e}\frac{d\vec{v}}{dt}e^{-t/\tau_e}$$

$$\Rightarrow \quad \frac{d}{dt}\left(\frac{d\vec{v}}{dt}e^{-t/\tau_e}\right) = -\frac{1}{m\tau_e}\vec{F}(t)e^{-t/\tau_e}$$

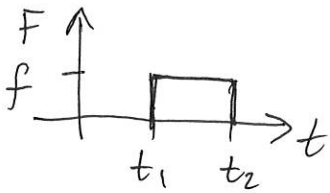
$$\Rightarrow \int_{-\infty}^{t} dt' \frac{d}{dt'}\left(\frac{d\vec{v}}{dt'}e^{-t'/\tau_e}\right) = \frac{d\vec{v}}{dt}e^{-t/\tau_e} - \left(\frac{d\vec{v}}{dt}e^{-t/\tau_e}\right)\Bigg|_{t=-\infty}$$

$$= -\frac{1}{m\tau_e}\int_{-\infty}^{t} dt' \, \vec{F}(t')e^{-t'/\tau_e}$$

For a particle at rest at $t=-\infty$ this term is absent

$$\frac{d\vec{v}(t)}{dt} = -\frac{1}{m\tau_e}\int_{-\infty}^{t} dt' \, e^{(t-t')/\tau_e}\vec{F}(t')$$

Consider $\vec{F}(t)$ to be a pulse of finite length



then $\frac{d\vec{v}}{dt} = -\frac{1}{m\tau_c}\int_{t_1}^{t_2} dt'\, \vec{f}\, e^{\frac{t-t'}{\tau_c}} =$

$$= -\frac{\vec{f}}{m} e^{\frac{t}{\tau_c}} \int_{\frac{t_1}{\tau_c}}^{t_2/\tau_c} d\left(\frac{t'}{\tau_c}\right) e^{-\frac{t'}{\tau_c}} =$$

$$= \frac{\vec{f}}{m} e^{\frac{t}{\tau_c}} \int d\left(e^{-t'/\tau_c}\right) = \frac{\vec{f}}{m} e^{\frac{t}{\tau_c}}\left(e^{-\frac{t_2}{\tau_2}} - e^{-\frac{t_1}{\tau_c}}\right)$$

$$\frac{d\vec{v}}{dt} = \frac{\vec{f}}{m}\left(e^{\frac{t-t_2}{\tau_c}} - e^{\frac{t-t_1}{\tau_c}}\right) \neq 0$$ even after time the pulse ends.

$\Rightarrow$ runaway solution is general feature of this equation

It only makes sense perturbatively when $\propto \frac{d^2\vec{v}}{dt^2}$ is considered to be small correction to regular motion

This means applicable only when radiation energy loss $\ll$ kinetic energy of the particle.

Conditions of applicability

Really, we should think of $\vec{F}_{rad}$ as a small correction.
For particle moving in external field (non-relativistically)

$$m\dot{\vec{v}} = q\vec{E} + q\,\vec{v}\times\vec{B} + \underbrace{\frac{\mu_0 q^2}{6\pi c}\ddot{\vec{v}}}_{\text{perturbation}}$$

$$\ddot{\vec{v}} \approx \frac{q}{m}\dot{\vec{E}} + \frac{q}{m}\dot{\vec{v}}\times\vec{B} = \frac{q}{m}\dot{\vec{E}} + \left(\frac{q}{m}\right)^2 \vec{E}\times\vec{B}$$

$$\dot{\vec{v}} = \frac{q}{m}\vec{E}$$

for motion with small velocity!

$$\Rightarrow \vec{F}_{rad} = \frac{\mu_0 q^2}{6\pi c}\left(\frac{q}{m}\dot{\vec{E}} + \left(\frac{q}{m}\right)^2 \vec{E}\times\vec{B}\right)$$

For oscillatory motion $\dot{\vec{E}} \sim \omega\vec{E}$

$$\Rightarrow 1^{st}\text{ term} = \frac{\mu_0 q^3 \omega}{6\pi c m}\vec{E}$$

$$2^{nd} = \frac{\mu_0 q^4}{6\pi c m^2}\vec{E}\times\vec{B}$$

F_{rad} should be small correction to the Coulomb/Lorentz forces

$\Rightarrow F_{rad} \ll$ Coulomb force $= qE$

$$\Rightarrow \frac{\mu_0 q^2}{cm}\omega \ll 1 \Rightarrow \underbrace{\frac{c}{\omega}}_{\lambda} \gg \frac{\mu_0 q^2}{m} = \frac{q^2}{mc^2\varepsilon_0} \sim R_0$$

$\Rightarrow$ radiation friction small if wavelength $\lambda \gg R_0$

Radius of electron $\frac{e^2}{4\pi\varepsilon_0 R_0} \sim mc^2 \Rightarrow R_0 \sim \frac{e^2}{4\pi\varepsilon_0 c^2 m}$

Free electron in EM wave $\vec{E}(t) = \vec{E}_0 e^{-i\omega t}$

$$m \frac{d\vec{v}}{dt} = q\vec{E}_0 e^{-i\omega t} + \alpha \frac{d^2\vec{v}}{dt^2} \qquad \alpha = \frac{\mu_0 q^2}{6\pi c}$$

let $\vec{v}(t) = \vec{v}_0 e^{-i\omega t}$

$\boxed{\text{Thomson scattering}}$

$$-m\omega i \vec{V}_0 = q\vec{E}_0 + \alpha(-i\omega)^2 \vec{V}_0$$

$$(\alpha\omega^2 - im\omega)\vec{V}_0 = q\vec{E}_0$$

$$\vec{v}(t) = \frac{q\vec{E}_0 e^{-i\omega t}}{-i\omega m + \alpha\omega^2}$$

The electron radiates at a rate $\frac{dP}{d\Omega} = \frac{\mu_0}{16\pi^2 c}(\ddot{\vec{d}} \times \hat{n})^2 = \frac{\mu_0 q^2}{16\pi^2 c}(\vec{a} \times \hat{n})^2$

$$\vec{a}(t) = \frac{-i\omega q \vec{E}_0 e^{-i\omega t}}{-i\omega m\left(1 + i\frac{\alpha\omega^2}{\omega m}\right)} = \frac{q\vec{E}_0 e^{-i\omega t}}{m(1 + i\omega\tau_e)}$$

where $\tau_e = \frac{\alpha}{m} = \frac{\mu_0 q^2}{6\pi m_e c} = \frac{q^2}{6\pi\varepsilon_0 c^3 m_e} \approx 10^{-23}\,s$

Physical observables are <u>real</u>

$\Rightarrow$ physical $\vec{a}$ is given by real part of

$$\vec{a}(t) = \mathrm{Re}\left(\frac{q\vec{E}_0 e^{-i\omega t}}{m(1+i\omega\tau_e)}\right) = \mathrm{Re}\left(\frac{q}{m}\frac{\vec{E}_0 e^{-i\omega t}(1-i\omega\tau_e)}{1+(\omega\tau_e)^2}\right)$$

$$= \frac{q}{m(1+\omega^2\tau_e^2)}\mathrm{Re}\left\{(\vec{E}_0' + i\vec{E}_0'')\underbrace{(\cos\omega t - i\sin\omega t)(1-i\omega\tau_e)}_{\cos\omega t - \omega\tau_e\sin\omega t - i(\omega\tau_e\cos\omega t + \sin\omega t)}\right\}$$

here $\vec{E}_0 = \mathrm{Re}(\vec{E}_0) + i\,\mathrm{Im}(\vec{E}_0)$

$= \vec{E}_0' + i\vec{E}_0''$

$$\vec{a}(t) = \frac{q}{m(1+\omega^2\tau_c^2)} \left\{ \vec{E}_0' (\cos\omega t - \omega\tau_c \sin\omega t) + \vec{E}_0'' (\omega\tau_c \cos\omega t + \sin\omega t) \right\}$$

$$\Rightarrow \frac{dP}{d\Omega} = \frac{\mu_0 q^2}{16\pi^2 c} \left(\frac{q}{m(1+\omega^2\tau_c^2)} \right)^2 \left[\cos\varphi (\cos\omega t - \omega\tau_c \sin\omega t) + \sin\varphi (\omega\tau_c \cos\omega t + \sin\omega t) \right]^2 \cdot \underbrace{(\vec{E}_R \times \hat{n})^2}_{E_R^2 \sin^2\theta}$$

Parameterize $\vec{E}_0' = \vec{E}_R \cos\varphi$

$\vec{E}_0' = \vec{E}_R \sin\varphi$

so that amplitude $\vec{E}_0 = \vec{E}_R e^{i\varphi}$

and $\vec{E}_R$ is <u>real</u> vector

Now we need to average over the period $T = 2\pi/\omega$

$$\frac{1}{T}\int_0^T dt \ldots = \int_0^{2\pi} \frac{d(\omega t)}{2\pi} \ldots$$

let $x = \omega t$ and use $\int_0^{2\pi} dx \cos^2 x = \int_0^{2\pi} dx \sin^2 x = \pi$, $\int_0^{2\pi} dx \cos x \sin x = 0$.

$$\frac{1}{2\pi}\int_0^{2\pi} dx \left[\cos^2\varphi (\cos x - \omega\tau_c \sin x)^2 + \sin^2\varphi (\omega\tau_c \cos x + \sin x)^2 + 2\cos\varphi \sin\varphi (\cos x - \omega\tau_c \sin x)(\omega\tau_c \cos x + \sin x) \right]$$

$$= \frac{\pi}{2\pi} \left[\cos^2\varphi (1 + (\omega\tau_c)^2) + \sin^2\varphi ((\omega\tau_c)^2 + 1) + 2\cos\varphi \sin\varphi \cdot (\omega\tau_c - \omega\tau_c) \right]$$

$$= \frac{1}{2}(1 + \omega^2\tau_c^2)$$ (irrespective of the phase φ, as it should be)

$$\Rightarrow \text{averaged } \overline{\frac{dP}{d\Omega}} = \frac{\mu_0 q^4 \quad E_R^2 \sin^2\theta}{32\pi^2 c\, m^2 (1+\omega^2\tau_c^2)}$$

Total scattered power

$$\overline{P} = \int d\Omega \frac{dP}{d\Omega} = \underbrace{2\pi \int_0^\pi d\theta \sin\theta \sin^2\theta}_{\frac{8\pi}{3}} \frac{\mu_0 q^4 E_R^2}{32\pi^2 m^2 c\,(1+\omega^2\tau_c^2)}$$

We compare it to the intensity of incoming EM wave

$$\vec{S} = \frac{1}{\mu_0} \vec{E}\times\vec{B} = \frac{1}{\mu_0 c} \vec{E}^2(t)\,\hat{n}$$

which too needs to be averaged over the period T

$$\overline{\vec{S}} = \frac{\hat{n}}{\mu_0 c} \int_0^{2\pi} \frac{d(\omega t)}{2\pi} \underbrace{\frac{1}{2}\left(\vec{E}_0 e^{-i\omega t} + \vec{E}_0^* e^{i\omega t}\right)}_{\text{physical el. field}} \cdot \frac{1}{2}\left(\vec{E}_0 e^{-i\omega t} + \vec{E}_0^* e^{i\omega t}\right)$$

$$= \frac{\hat{n}}{\mu_0 c} \frac{1}{4} \int_0^{2\pi} \frac{d(\omega t)}{2\pi} \Big(E_0^2 e^{-i2\omega t} + (E_0^*)^2 e^{i2\omega t} + 2\,\vec{E}_0\cdot\vec{E}_0^* \Big) = \hat{n}\,\frac{|\vec{E}_0|^2}{2\mu_0 c}$$

Note that $|\vec{E}_0|^2 = E_R^2$

$$\Rightarrow \frac{\overline{P}}{|\overline{\vec{S}}|} = \frac{8\pi}{3} \cdot \frac{2\mu_0 c \cdot \mu_0 q^4}{32\pi^2 m^2 c\,(1+\omega^2\tau_c^2)} =$$

$$= \frac{\mu_0^2 q^4}{6\pi m^2} \frac{1}{1+\omega^2\tau_c^2} = \frac{\sigma_T}{1+\omega^2\tau_c^2}$$

$$\sigma_T = \frac{\mu_0^2 q_e^4}{6\pi m_e^2} = \frac{e^4}{6\pi m_e^2 \varepsilon_0^2 c^4}$$ = Thomson scattering cross section

Observe that for $\omega \ll \frac{1}{\tau_e}$ $\frac{\bar{P}}{S} = \sigma_T$ is frequency independent.

At large frequences $\omega \gg \frac{1}{\tau_e}$ the scattering vanishes.

But such frequences are too large for classical physics: $\omega = \frac{1}{\tau_e} \Rightarrow$ photon energy $\mathcal{E}_{ph} = \hbar\omega = \frac{\hbar}{\tau_e}$

Let's compare $$\frac{\mathcal{E}_{ph}}{m_e c^2} = \frac{6\pi \varepsilon_0 c^3 m_e \hbar}{q_e^2 m_e c^2} = \frac{6\pi \varepsilon_0 c \hbar}{q_e^2} = \frac{6}{4} \frac{4\pi\varepsilon_0 \hbar c}{e^2}$$

But Fine structure constant $= \frac{e^2}{4\pi\varepsilon_0 \hbar c} = \frac{1}{137}$

$$\Rightarrow \frac{\mathcal{E}_{ph}}{mc^2} = \frac{3}{2} \cdot 137 = 205 \gg 1.$$

$\Rightarrow$ At $\omega \sim \frac{1}{\tau_e}$ photon energy is 200 times greater than electron's rest energy

$\Rightarrow$ classical treatment is not adequate.

Another example:

charged oscillator

$$m\ddot{\vec{x}} = -k\vec{x} + \frac{\mu_0 q^2}{6\pi c}\dddot{\vec{x}}$$

$$\ddot{x} = -\frac{k}{m}x + \frac{\mu_0 q^2}{6\pi mc}\dddot{x}$$

small correction to a periodic motion

But $\dddot{x} \approx -\frac{k}{m}\dot{x} = -\omega_0^2\dot{x}$

$$\Rightarrow \quad \ddot{x} = -\omega^2 x - \frac{\mu_0 q^2}{6\pi mc}\omega_0^2\dot{x}$$

$$\ddot{x} + \underbrace{\alpha\dot{x}}_{\text{frictional force proportional to } \vec{v}!} + \omega_0^2 x = 0$$

Add external field $q\vec{E}$ on the right-hand side

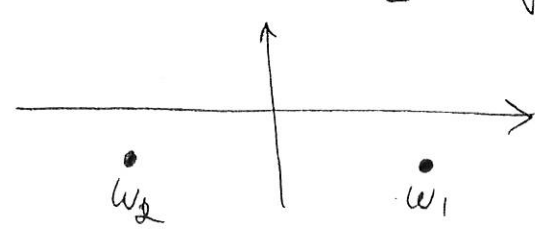
$\Rightarrow$ problem 1 from HW01, with $\gamma = \alpha \cdot m = \frac{\mu_0 q^2 \omega_0^2}{6\pi c}$

$$x(t) = -\frac{q}{m}\int \frac{d\omega}{2\pi}\frac{\vec{E}_\omega e^{-i\omega t}}{\omega^2 + i\gamma\omega - \omega_0^2}$$

Thus friction $\gamma = m\alpha$ gives us a rule for selecting contour of integration over ω.

$$\omega_{1,2} = -i\frac{\gamma}{2} \pm \sqrt{\omega_0^2 - \left(\frac{\gamma}{2}\right)^2}$$

$\mathrm{Im}(\omega_{1,2}) < 0$



"The end of classical physics"

<u>Hydrogen atom classically:</u>

$$E = \frac{1}{2} m v^2 - \frac{e^2}{4\pi\varepsilon_0 r} = -\frac{e^2}{8\pi\varepsilon_0 r} \qquad r_0 = 0.5\,\text{Å} \sim 10^{-10}\,\text{m}$$

$$F = \frac{e^2}{4\pi\varepsilon_0 r^2} = m \underbrace{\frac{v^2}{r}}_{a} \Rightarrow m v^2 = \frac{e^2}{4\pi\varepsilon_0 r}$$

$$P = \frac{\mu_0 e^2}{6\pi c} a^2 = \frac{\mu_0 e^2}{6\pi c}\left(\frac{v^2}{r}\right)^2 = \frac{\mu_0 e^2}{6\pi c}\left(\frac{e^2}{4\pi\varepsilon_0 m r^2}\right)^2$$

But $$P = -\frac{dE}{dt} = \frac{d}{dt}\left(\frac{e^2}{8\pi\varepsilon_0 r}\right) = -\frac{e^2}{8\pi\varepsilon_0 r^2}\frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = -\frac{8\pi\varepsilon_0}{e^2}\frac{\mu_0 e^2}{6\pi c}\left(\frac{e^2}{4\pi\varepsilon_0 m r}\right)^2 =$$

$$= -\frac{1}{3c}\left(\frac{e^2}{2\pi\varepsilon_0 m c}\right)^2 \frac{1}{r^2}$$

$$T = \int dt = -3c\left(\frac{2\pi\varepsilon_0 m c}{e^2}\right)^2 \int_{r_0}^{0} r^2 dr = c\left(\frac{2\pi\varepsilon_0 m c}{e^2}\right)^2 r_0^3$$

$$T \simeq 3\cdot 10^8 \left[\frac{2\pi \; 8.85\cdot 10^{-12}\cdot 9\cdot 10^{-31}\cdot 3\cdot 10^8}{(1.6\cdot 10^{-19})^2}\right]^2 (10^{-10})^3$$

$$= 3\,\pi^2\,(9^2)^2\,3^2\,10^{8-30}\,(10^3)^2$$

$$\sim 3\cdot 9\cdot 9\cdot 9^4\,10^{-16} \sim \underline{\underline{10^{-10}\,\text{s}}}$$

$\Rightarrow$ there are <u>no atoms</u>!

For a particle of mass m

$$\Delta p \cdot \Delta x \geq \frac{\hbar}{2}$$

$$\Delta x_{min} < \frac{\hbar}{\Delta p_{max}} = \frac{\hbar}{mc} = \text{"quantum radius"} \; R_q$$

Classical radius of charged particle $\frac{e^2}{4\pi\varepsilon_0 R_0} = mc^2$

$$R_0 = \frac{e^2}{4\pi\varepsilon_0 mc^2}$$

$$\frac{R_0}{R_q} = \frac{e^2}{4\pi\varepsilon_0 mc^2} \frac{mc}{\hbar} = \frac{e^2}{4\pi\varepsilon_0 \hbar c} = \frac{1}{137}$$

$$\Rightarrow R_0 \ll R_q$$

Quantum effects begin well before classical EM breaks down.